Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 4: The Capacitor

M. Mert Ankaralı

Colour code used in these notes and in the slides			
charge q	current i	voltage v	power p, energy w

1. An element that stores energy

Every element so far — sources, diodes, the LTI resistor — has been a *resistor* in the sense of Lecture 3: its characteristic is a curve in the v–i plane, so the present current is fixed by the present voltage and nothing else. Look at what that does to energy. In a resistor $p = Ri^2 \geq 0$ at every instant: the element **dissipates** the energy it is given, turning it into heat and letting it go. It flows one way only, and none of it ever comes back.

Today's element dissipates nothing. Energy goes in, *stays* in, and comes back out later when the circuit asks for it — all of it. A capacitor is an **ideal energy-storage element**, and that is also why it remembers: what it does now depends on everything that has been put into it.

Lecture 3
a curve in the v–i plane
energy in → dissipated as heat

⟶

Today
a curve in the v–q plane
energy in → stored → back out

1.1. Why energy is the bigger idea

Charge is the natural bookkeeping variable inside an electrical circuit, and the next few pages are written in terms of it. But charge only means something electrically. Energy means the same thing everywhere — and an ideal energy-storage element is a shape that every physical domain has:

capacitor	**moving mass**	**flywheel**
voltage v	velocity u	angular velocity ω
$\frac{1}{2}Cv^2$	$\frac{1}{2}mu^2$	$\frac{1}{2}J\omega^2$

Each one takes energy in, holds it, and gives it back — and in each the stored energy is half a constant times the square of one variable. The resistor's mechanical counterpart is *viscous friction*: it too takes energy in, and there the resemblance stops. Lecture 1 lined the domains up variable by variable; this is the same table read one level higher, in the one quantity they all share.

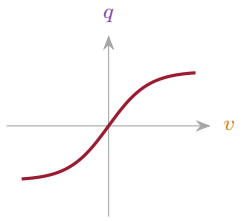
2. The capacitor: a curve in a different plane

Some elements store charge. For those, the useful pair of variables is not (v, i) but (v, q), where $q(t)$ is the charge that has entered the + terminal:

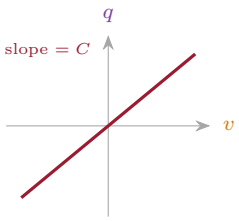
$$q(t) = q(t_0) + \int_{t_0}^{t} i(\tau)\, d\tau \qquad \Longleftrightarrow \qquad i = \frac{dq}{dt} \quad \text{(Lecture 1)}.$$

Capacitor

A two-terminal element is a **capacitor** if, at every instant, its (v, q) pairs lie on a curve in the v–q plane.



a capacitor in general



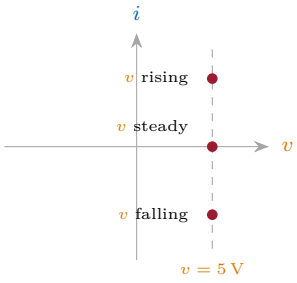
the LTI capacitor

A bent curve is a nonlinear capacitor; a varactor diode is one. In this course every capacitor is **linear and time-invariant**, so from here on the characteristic is the straight line.

Note what has *not* happened. There is of course a relation between v and i — we derive it two pages from here — but it is not a relation between their *values at one instant*. For a capacitor there is no function f or g with

$$i(t) = f\big(v(t)\big) \qquad \text{or} \qquad v(t) = g\big(i(t)\big).$$

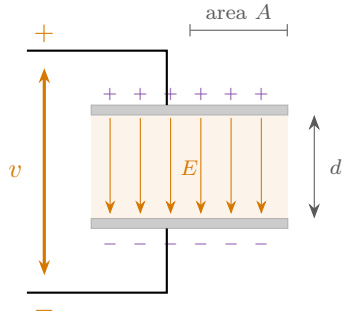
Fix the voltage at some value and the current is still free: it depends on how v is *changing*, not on what it is. So a capacitor is not a resistor, and it has no characteristic in the v–i plane at all.



One voltage, any current — three of them drawn here. A resistor's characteristic would have to pick exactly one point on this dashed line; a capacitor allows the whole line.

3. Where the charge sits: the parallel-plate capacitor

Two conducting plates of area A, a distance d apart, with an insulator of permittivity ε between them. Push charge $+q$ onto one plate and the same amount leaves the other, so that plate carries $-q$; the circuit outside never sees any net charge appear or vanish.



$$q = C\,v, \qquad C = \frac{\varepsilon A}{d}$$

Unit: the **farad**, F = C/V. Wide plates close together, with a material of high permittivity between them, give a large C.

One farad is enormous

Plates of $1\,\text{cm}^2$ separated by $0.1\,\text{mm}$ of air give

$$C = \frac{\varepsilon_0 A}{d} = \frac{(8.85\times10^{-12})(10^{-4})}{10^{-4}} \approx 8.9\,\text{pF}.$$

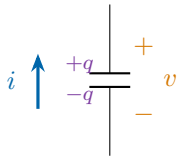
That is why practical values are quoted in pF, nF and μF, and why a 1 F supercapacitor has to roll an enormous area into a small can.

4. The LTI capacitor

LTI capacitor

The v–q characteristic is a fixed straight line through the origin, $q = C\,v$ with C constant. Differentiating and using $i = dq/dt$,

$$\boxed{i \;=\; C\,\frac{dv}{dt}} \qquad \text{or, integrated,} \qquad \boxed{v(t) = v(t_0) + \frac{1}{C}\int_{t_0}^{t} i(\tau)\,d\tau}$$



Two equations, two readings. **Differential**: the current is whatever the *rate of change* of voltage demands. **Integral**: the voltage is the running total of everything the current has done since t_0.

Four consequences, straight from the two boxes

- **At DC it is an open circuit.** v constant $\Rightarrow dv/dt = 0 \Rightarrow i = 0$, whatever the voltage is.
- **Voltage cannot jump** — not with a finite current. A step in v would need $dv/dt = \infty$.
- **It remembers.** $v(t)$ depends on the whole history of i, not on its present value. A resistor knows only now; a capacitor knows everything that has happened to it.
- **It holds energy**, and the amount is $w = \frac{1}{2}Cv^2$ — fixed by the present voltage alone. Section 6 shows why, and that every joule of it can be taken back out.

5. Two examples that show the two readings

Example 4.1 — drive it with a constant current

A current source pushes a constant $I = 1\,\text{mA}$ into $C = 1\,\mu\text{F}$, starting from $v(0) = 0$. The integral form gives

$$v(t) = \frac{1}{C}\int_0^t I\,d\tau = \frac{I}{C}\,t = \frac{10^{-3}}{10^{-6}}\,t = 1000\,t \quad [\text{V}].$$

A **ramp**: the voltage climbs at 1000 V/s, reaching 5 V after 5 ms.



constant current in



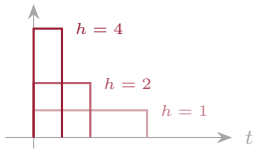
voltage ramps

The capacitor is **integrating** the current. Nothing about v can be read off i at one instant — only from the area under it.

In practice the ramp cannot go on for ever: the source will run out of voltage. Until it does, this is exactly how a sawtooth generator works.

5.1. A first look at the impulse

The next example needs a signal we have not met. Start with an ordinary rectangular pulse of height h and width $1/h$, beginning at $t = 0$:



area $= h \cdot \dfrac{1}{h} = 1$, whatever h is.

Let h grow: the pulse gets taller, the width $1/h$ shrinks towards zero, and the area never moves.

In the limit nothing is left that we could call a value — the pulse is zero everywhere except at $t = 0$, where it has run off to infinity — but the *area* is still exactly 1. That limit is the **unit impulse** $\delta(t)$.

Unit impulse $\delta(t)$

$\delta(t) = 0$ for every $t \neq 0$, and

$$\int_a^b \delta(\tau)\, d\tau = 1 \quad \text{for any } a < 0 < b.$$

It is not a function in the ordinary sense; it is defined by the area it carries. Scaling scales the area: $K\,\delta(t)$ has area K. We draw it as a vertical arrow, and the number written beside the arrow is its **area**, never a height.

Two facts we will use straight away. First, a current impulse $K\,\delta(t)$ delivers charge K in zero time, since charge is the area under the current. Second, the impulse is what you get by differentiating a step: if v jumps by V at $t = 0$, then $dv/dt = V\,\delta(t)$ — zero before, zero after, and area V at the jump.

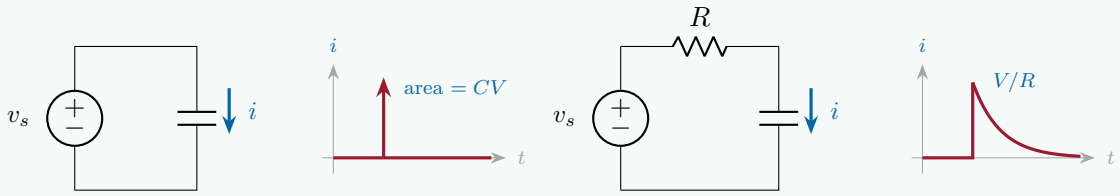
Example 4.2 — drive it with a constant voltage

Now hold $v = V$ constant. Then $dv/dt = 0$, so

$$i = C\frac{dv}{dt} = 0, \qquad q = C\,V \text{ sits on the plates and stays there.}$$

The capacitor has become an *open circuit*, and it stores $q = CV$ while it does so.

But how did the voltage get there? Compare two circuits, both driven by a source that steps from 0 to V at $t = 0$.



nothing in series — an impulse — any real circuit — tall, but finite

On the left there is nothing to limit the current. Differentiate the step as in Section 5.1:

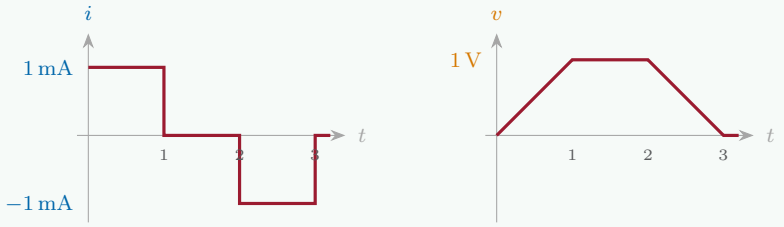
$$i(t) \;=\; C\frac{dv}{dt} \;=\; C\,V\,\delta(t),$$

an impulse of area CV — exactly the charge the capacitor must end up holding, delivered in zero time. The model is telling us that an ideal voltage step across an ideal capacitor is one idealisation too far.

On the right the resistance limits the current to V/R at the instant of the step, and from there it decays while the capacitor voltage rises smoothly to V. That is what really happens, and we will solve this circuit properly when we reach first-order transients.

Example 4.3 — reading a current waveform

The same $1\,\mu\text{F}$ capacitor, now fed the current below, starting from $v(0) = 0$. Between the corners i is constant, so v is a straight line whose slope is i/C; where $i = 0$ the voltage simply holds.



current in, ms voltage out, ms

Up at $1000\,\text{V/s}$ for a millisecond, flat for a millisecond, then down again — and the capacitor ends where it started, having given back exactly the charge it took. Read the two pictures the other way round and you have the differentiating view: the voltage's slope *is* the current, to within the factor C.

6. Power and energy in an LTI capacitor

$$p = v\,i = v\,C\,\frac{dv}{dt} = \frac{d}{dt}\left(\tfrac{1}{2}Cv^2\right) \quad\Longrightarrow\quad \boxed{w = \tfrac{1}{2}C\,v^2 = \frac{q^2}{2C}}$$

So the power is the rate of change of $\frac{1}{2}Cv^2$. That quantity is the **energy the capacitor holds**, and it depends only on the present voltage — not on how the capacitor got there.

	LTI resistor (Lecture 3)	**LTI capacitor**
terminal equation	$v = Ri$	$i = C\,dv/dt$
plane it lives in	v–i	v–q
memory	none	the whole past of i
power p	$Ri^2 \geq 0$ always	$\frac{d}{dt}(\frac{1}{2}Cv^2)$, either sign
energy	dissipated as heat, gone	stored, and all of it returnable
verdict	passive, *lossy*	passive, *lossless*

$w = \frac{1}{2}Cv^2 \geq 0$, so a capacitor can never give out more than it has been given: it is passive. But p goes negative whenever $|v|$ falls, and then the capacitor is handing energy back to the circuit. Nothing is lost on the way in or on the way out.

For the mechanically minded

Compare the terminal equation with Newton's second law:

$$i = C\,\frac{dv}{dt} \qquad\longleftrightarrow\qquad F = m\,\frac{du}{dt}.$$

Current plays the part of force, voltage the part of velocity, and capacitance the part of mass — the same pairing as the mechanical column of Lecture 1. The energies match too: $\frac{1}{2}Cv^2$ against $\frac{1}{2}mu^2$.

A heavy flywheel is a large capacitance: a finite force cannot change its speed instantly, just as a finite current cannot change a capacitor's voltage instantly.

Example 4.4 — how much energy is that?

A $100\,\mu$F capacitor charged to 12 V holds

$$w = \tfrac{1}{2}(10^{-4})(12)^2 = 7.2\,\text{mJ}.$$

Enough to light a small LED for a few seconds — and about what a camera flash unit dumps into the lamp in a millisecond, which is the point of using a capacitor rather than the battery directly.

Summary

- A capacitor is an **ideal energy-storage element**: it dissipates nothing, and everything put in can be taken back out. Every element of Lecture 3 did the opposite — it dissipated all of it.
- A **capacitor** is a curve in the v–q plane, not the v–i plane. It is not a resistor: there is no f with $i(t) = f(v(t))$, because the current follows how v is changing, not what it is.
- Parallel plates: $C = \varepsilon A/d$, in farads.
- The **LTI capacitor** has $q = Cv$, hence $i = C\,dv/dt$ and $v(t) = v(t_0) + \frac{1}{C}\int i\,d\tau$. Open circuit at DC; voltage cannot jump; it remembers.
- Constant current in gives a voltage **ramp**; a voltage **step** would demand $i = CV\delta(t)$, an **impulse** — the limit of a rectangle of height h and width $1/h$, defined by its area and not by any value. That is the model's way of saying that some series resistance is always there.
- $w = \frac{1}{2}Cv^2 = q^2/2C$: stored, not spent — and fixed by the present voltage, however the capacitor got there. Passive and *lossless*, unlike the resistor.

Next lecture

The inductor — the second ideal energy-storage element, and the mirror image of everything above: flux linkage and current in place of charge and voltage, $\frac{1}{2}Li^2$ in place of $\frac{1}{2}Cv^2$. With those two elements the course has all it needs to build circuits that do more than divide.